Sahaj Bhadja

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Summary

Cloud engineer experienced in **Python automation, AWS, Azure, GCP, Kubernetes, CI/CD, observability, distributed systems, and AI workloads**, building reliable infrastructure and production services with measurable performance gains.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Cloud Computing, Operating Systems, Database Systems, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Computer Networks, Software Engineering, Data Structures, Object-Oriented Programming

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Architected a multi-tenant platform on **Azure SQL and Blob Storage** with row-level isolation, indexed queries, and service monitoring, reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Reworked asynchronous Python orchestration with **explicit job states, concurrent workers, failure isolation, and operational diagnostics**, tripling throughput for 1K+ AI jobs while sustaining 99%+ reliability across production releases.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Operated Flask services and MySQL workloads on AWS RDS, supporting 10K+ daily requests through **reliable cloud monitoring, secure ingestion, resilient processing, production debugging, and repeatable CI/CD-backed deployments**.
- Developed multithreaded Python processing for PDF, DOCX, ZIP, OCR, and multimodal workloads, reducing turnaround 60% through **bounded concurrency, fault tolerance, resource validation, automated testing, and performance tuning**.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built reproducible Python evaluation pipelines processing 100K+ simulated driving trajectories, improving planning 18% while applying **batch execution, resource profiling, instrumentation, controlled benchmarking, and repeatable AI workload validation**.
- Implemented a spatial-temporal encoder combining HD-map geometry and ego trajectories with multi-head attention, improving closed-loop stability 15% through **experiment monitoring, failure analysis, performance measurement, and documented research operations**.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Designed distributed data pipelines using **Python, AWS Glue, S3, SQL, and asynchronous processing**, improving response times 30% while supporting reliable cross-platform data movement, monitoring, and operational troubleshooting.
- Automated 20+ unit and integration suites across Windows and Linux virtual machines using PyTest, Mockito, and Jenkins, increasing pipeline efficiency 68% through **repeatable builds, regression isolation, and diagnostics**.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built eight modular services and real-time WebSocket flows synchronizing 2M+ records through **Azure SQL, REST APIs, event-driven updates, shared schemas, authentication, and failure monitoring** across distributed application components.
- Automated Docker and Jenkins deployments, reducing release time 50% while applying **Linux containers, Sentry observability, structured logs, integration testing, rollback-ready builds, and production troubleshooting** across customer-facing services.

Skills

Cloud: GCP, AWS, Azure, GKE, EKS, Kubernetes, Bedrock, Vertex AI, IAM, SSO
Infrastructure: Terraform, CloudFormation, CDK, Ansible, AWX, Docker, Networking, Linux, HPC, Distributed Systems
Operations: Datadog, CloudWatch, GCP Monitoring, Prometheus, Logging, Observability, Incident Response, RCA, FinOps, Cost Optimization
Engineering: Python, Bash, SQL, Git, Jenkins, GitHub Actions, CI/CD, PyTest, Unit Testing, Debugging

Projects

Multi-Cloud Landing Zone | *Terraform, GCP, AWS, GKE, IAM, Cloud Monitoring, GitHub Actions*

- Built a **multi-cloud landing zone** using Terraform modules for networks, IAM, GKE clusters, storage, budgets, and centralized logs, enforcing policy checks through automated CI/CD plans and deployments.

AI Workload Operations Platform | *Python, Vertex AI, AWS Bedrock, Kubernetes, Datadog, Ansible*

- Developed an **AI workload operations platform** that provisions inference services, tracks latency and cost, configures alerts, automates recovery runbooks, and compares capacity across GCP and AWS deployment targets.

Linguistix Cloud Platform | *Python, AWS, Docker, CI/CD, Serverless Architecture*

- Engineered an AWS language platform with **serverless processing, Dockerized services, automated deployments, and scalable storage**, supporting reproducible environments while improving delivery reliability through continuous integration and operational monitoring.